

Daja 20W Blue Diode Laser Cutting Guide

Fixed settings for maximum material variety across the 500 to 11000 mm/min speed ramp.

Filter by Material, Speed, or Mode...

Layer	Material (Type & Thickness)	Mode	Speed (mm/min)	Power (%)	Passes	Key Application / Material Variety
L00	12mm Pine / 8mm Bamboo / 6mm Red Oak / 10mm Poplar / 8mm MDF	Line	500	95	8	MAXIMUM CUT: Used for the thickest and most challenging woods/boards.
L01	6mm Baltic Birch / 5mm MDF / 4mm Hardwood / 5mm Walnut Ply / 3mm UHMW	Line	750	90	5	THICK CUT: General clean cutting of standard thick crafting materials.
L02	4mm Cherry Ply / 3mm White Acrylic / 2mm ABS / 4mm Mahogany / 3mm HDPE	Line	1000	85	3	MEDIUM CUT: Best for clean cuts on plastics, denser 2mm-4mm woods.
L03	3mm Basswood / 2mm Leather / 1mm Polycarbonate	Line	1250	100	2	THIN CUT: Good for solid soft woods, leather, and thin industrial plastics.
L04	3mm Black Acrylic / 2mm Delrin / 1mm HDPE / 2mm Cast Acrylic / 3mm Styrene	Line	1500	95	3	PLASTIC CUT: High passes for clean melt/cut on high-quality plastics.
L05	3mm Thick Bamboo / 2mm Hardboard / 1mm Fiberglass / 2mm Laminated Wood / 3mm particleboard	Line	1750	100	4	FIBROUS CUT: For materials with high glue content or inconsistent density.
L06	2mm Suede / 1.5mm Natural Rubber / 3mm Cork Sheet / 1mm EVA Foam (Dense) / 2mm Heavy Felt	Line	2000	70	1	FLEXIBLE CUT: Reduced power prevents charring/melting on flexible, organic materials.
L07	5mm EVA Foam / 3mm EPP Foam / 2mm Thick Cardboard / 4mm Packaging Foam / 3mm Craft Felt	Line	2250	40	1	FOAM CUT: Low power for clean cuts through various types of foam.
L08	1mm Cherry Veneer / 1.5mm Wool Felt / 0.5mm Gasket Paper / 1mm Basswood Veneer / Thin Leatherette	Line	2500	90	1	THIN WOOD/FABRIC CUT: Single pass, high power for fast production of thin materials.
L09	ENGRAVE: Deep Wood / Dense Cork / Black Paint / Slate / Heavy Cardboard Etch	Fill	2750	75	1	DEEP ENGRAVE (FILL): Slowest fill for maximum depth and material removal.
L10	5mm Clear Acrylic (Score) / 4mm White PVC (Score) / 3mm HDPE (Score) / 2mm Polycarbonate (Score)	Line	3000	90	2	PLASTIC SCORE: Two passes for deep scribing on plastics that don't cut well.
L11	1mm Chipboard / 0.5mm Mat Board / 1mm Thin Craft Ply / 0.75mm Museum Board / 0.5mm Art Paper	Line	3250	60	1	CARD CUT: Moderate power for paper products to reduce char.
L12	ENGRAVE: Large Area Wood / Rough Stippling / Fast Cork Etch / Fast Plywood Labeling / Fast ABS Marking	Offset Fill	3500	90	1	OFFSET FILL: Fast, high-power rough etching for large areas or quick texture.
L13	2mm Basswood Ply (Score) / Masking Paper / Thin Veneer Score / 1mm Thin Cardboard (Score)	Line	3750	50	1	LIGHT SCORE: Good for reference lines, shallow scoring, or cutting masking.
L14	4mm Solid Hardwood (Deep Score) / 3mm Delrin (Deep Score) / 2mm Thick Copper Laminate (Score)	Line	4000	100	1	DEEP SCORE/MARK: High energy at a single pass for marking dense/hard surfaces.
L15	2mm Polyester Fabric / 1mm Dark Neoprene / 0.5mm Vinyl (Thick) / 1.5mm Synthetic Leather	Line	4250	30	1	SYNTHETIC CUT: Low power for heat-sensitive synthetic fabrics/films.
L16	ENGRAVE: Thin Foam / Light Wood Shading / Fast Texture / Fast Felt Engrave / Cardboard Shading	Offset Fill	4500	50	1	OFFSET FILL: Fast, light fill for surface texture on soft materials (cork, foam).
L17	ENGRAVE: Detailed Wood Shading / Cermark on Steel / Glass / Ceramic	Fill	5000	70	1	DETAIL ENGRAVE (FILL): Mid-speed for good grayscale/detail on wood.
L18	ENGRAVE: Slate / Ceramic Tile / Anodized Aluminum / Painted Metal / Dark Granite	Fill	5500	90	1	COATING REMOVAL: High power for effective removal of paint/anodizing.
L19	SCORE: Plywood Fold Lines / Fine Acrylic Scribe / Thin Veneer Mark / 0.5mm Styrene Score	Line	6000	40	1	FINE SCRIBE: Use for precise fold lines, light decoration, or reference.
L20	SCORE: Faint Leather Mark / Light Paper Crease / Alignment Dots / Thin Card Stock Mark	Line	6500	25	1	LIGHTEST MARK: Minimal power for barely visible marks or alignment checks.
L21	1mm Balsa / 0.5mm Cardboard / 0.5mm Craft Foam / 0.3mm Construction Paper / 0.5mm Chipboard	Line	7000	70	1	BALSA/THIN CUT: Max speed for extremely soft or light paper materials.
L22	0.5mm Gasket Paper / 0.2mm Mylar Film / 0.1mm Thin Leather / 0.3mm Thin Rubber	Line	7500	50	1	FILM/PAPER CUT: Fast cutting of very thin materials.
L23	ENGRAVE: Fast Logo Etch / Light Text on Wood / Thin Film Etch / Fast Paper Labeling	Fill	8000	60	1	FAST FILL: Quick fill for basic logos and text where deep etching isn't needed.
L24	1mm PET Sheet / 0.5mm Polypropylene / 0.3mm Polyethylene / 0.2mm Craft Vinyl (Thick)	Line	8500	80	1	THIN PLASTIC CUT: Fast cutting of thin plastic sheets and films.
L25	ENGRAVE: Fast Delrin Mark / 1mm HDPE Etch / Quick Labeling / Fast Thin Plywood Etch	Offset Fill	9000	75	1	OFFSET FILL: Fastest option for quick, high-contrast marking of plastic parts.
L26	0.5mm Thin Paper / 0.1mm Sticker Paper / Masking Tape / 0.05mm Heat Transfer Vinyl	Line	9500	20	1	DELICATE CUT: Very low power for vinyl/sticker paper (fast on backside).
L27	ENGRAVE: Coated Metal / Anodized Aluminum (Fastest) / Glass Etch / Cermark on Chrome	Fill	10000	100	1	MAX SPEED ENGRAVE: Full power at high speed for maximum color/coating removal.
L28	0.1mm Vinyl / 0.1mm Thin Film / Temporary Stencils / Paper Towel / Tissue Paper	Line	10500	10	1	LOWEST POWER CUT: Absolute minimum power for cutting delicate films/stencils.
L29	Anything	Line	7000	10	1	Framing
T1	Tool Layer 1: Cut Reference/Outlines	Line	100	1	1	Use for non-marking positioning guides.
T2	Tool Layer 2: Alignment/Ruler	Fill	4500	20	1	Use for fast, non-damaging tool marks or temporary rulers.

 Gemini API Key (External Use Only)

Please remember to turn on ProtonVPN before asking questions.

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Paste your active Gemini API Key here...

Ask the Daja Laser Expert (AI Consult)

Have a quick question? Ask the AI what layer or setting you should use for a material based on the guide above.

e.g., What layer should I use for cutting 5mm MDF?

Get Recommendation

Your AI recommendation will appear here.